

EXPERIMNET 1

1. AIM:

To perform data acquisition and profiling using Microsoft Power BI.

2. OBJECTIVES:

- To understand the basic concepts of Business Intelligence, data acquisition, and data profiling.
- To acquire data from multiple heterogeneous sources such as CSV, Excel, and SQL Server.
- To import and integrate different datasets into Microsoft Power BI.
- To examine the structure, data types, completeness, uniqueness, and validity of the datasets.
- To examine the structure, data types, completeness, uniqueness, and validity of the datasets.
- To use Power Query Editor's Column Quality, Column Distribution, and Column Profile features for data profiling.
- To document the identified data quality issues and suggest suitable corrective actions before further analysis.

3. TOOLS / SOFTWARE REQUIRED:

- Microsoft Power BI Desktop
- Sample datasets: Global Superstore 2018

4. DATASET USED: Global Superstore 2018

5. THEORY:

Business Intelligence (BI) **Business Intelligence (BI)** refers to the technologies and methods used to collect, process, analyze, and present data to support effective decision-making. Microsoft Power BI is a Business Intelligence tool that allows users to import data, transform it, analyze it, and create interactive reports and dashboards.

Data Acquisition is the process of collecting and importing data from a source into an analytical environment. Power BI supports different data sources, including CSV files. A CSV file stores tabular data in text format, with values generally separated by commas. Power BI can import a CSV file using the Text/CSV option under Get Data.

Data Profiling is the process of examining a dataset to understand its structure, values, and quality before performing further analysis. It helps identify missing values, errors, duplicate or unique values, incorrect data types, and unusual data patterns.

In Power BI, data profiling is performed mainly through the **Power Query Editor**. It provides three important profiling features:

Column Distribution:

Column Distribution displays the frequency and distribution of values in a column. It also provides

information about distinct and unique values, which helps in identifying repeated or uncommon values.

Column Profile:

Column Profile provides detailed information about a selected column, including statistics and value distribution. It helps in understanding the characteristics and quality of individual columns.

For the **Global Superstore 2018.csv** dataset, these profiling features can be used to examine fields such as order information, customer details, product information, sales, quantity, profit, and dates. The profiling results help identify data quality issues and prepare the dataset for further analysis and visualization in Power BI.

6. PROCEDURE:

Part A: Software Setup and Data Acquisition

Step 1: Install the required software.

- Download and install Microsoft Power BI Desktop.

Step 2: Launch Power BI Desktop and import the CSV file.

Home > Get Data > Text/CSV

Browse to Global Superstore 2018.xlsx > Load

Table: Orders (51,290 rows)

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Postal Co
27884	ID-2017-DP131657-42742	Saturday, January 7, 2017	Wednesday, January 11, 2017	Standard Class	DP-131657	David Philippe	Consumer	
27883	ID-2017-DP131657-42742	Saturday, January 7, 2017	Wednesday, January 11, 2017	Standard Class	DP-131657	David Philippe	Consumer	
26896	ID-2014-FP143207-41945	Sunday, November 2, 2014	Saturday, November 8, 2014	Standard Class	FP-143207	Frank Preis	Consumer	
30039	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
30040	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
30041	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
27444	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27443	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27445	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27446	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
25656	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25655	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25657	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25658	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
27561	ID-2015-CS124607-42189	Saturday, July 4, 2015	Wednesday, July 8, 2015	Standard Class	CS-124607	Chuck Sachs	Consumer	
28358	IN-2016-JF154907-42399	Saturday, January 30, 2016	Thursday, February 4, 2016	Standard Class	JF-154907	Jeremy Farry	Consumer	
24819	ID-2016-JF153557-42379	Sunday, January 10, 2016	Thursday, January 14, 2016	Standard Class	JF-153557	Jay Fein	Consumer	
24818	ID-2016-JF153557-42379	Sunday, January 10, 2016	Thursday, January 14, 2016	Standard Class	JF-153557	Jay Fein	Consumer	
22411	ID-2016-DJ136307-42524	Friday, June 3, 2016	Thursday, June 9, 2016	Standard Class	DJ-136307	Doug Jacobs	Consumer	
22118	IN-2016-LS172307-42490	Saturday, April 30, 2016	Friday, May 6, 2016	Standard Class	LS-172307	Lycoris Saunders	Consumer	

Part B: Data Profiling in Power Query Editor

Step 3: Open Power Query Editor.

24 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON FRIDAY, JULY 31, 2026

Home > Transform Data

Step 4: Enable data profiling features.

View tab > Tick Column Quality

View tab > Tick Column Distribution

24 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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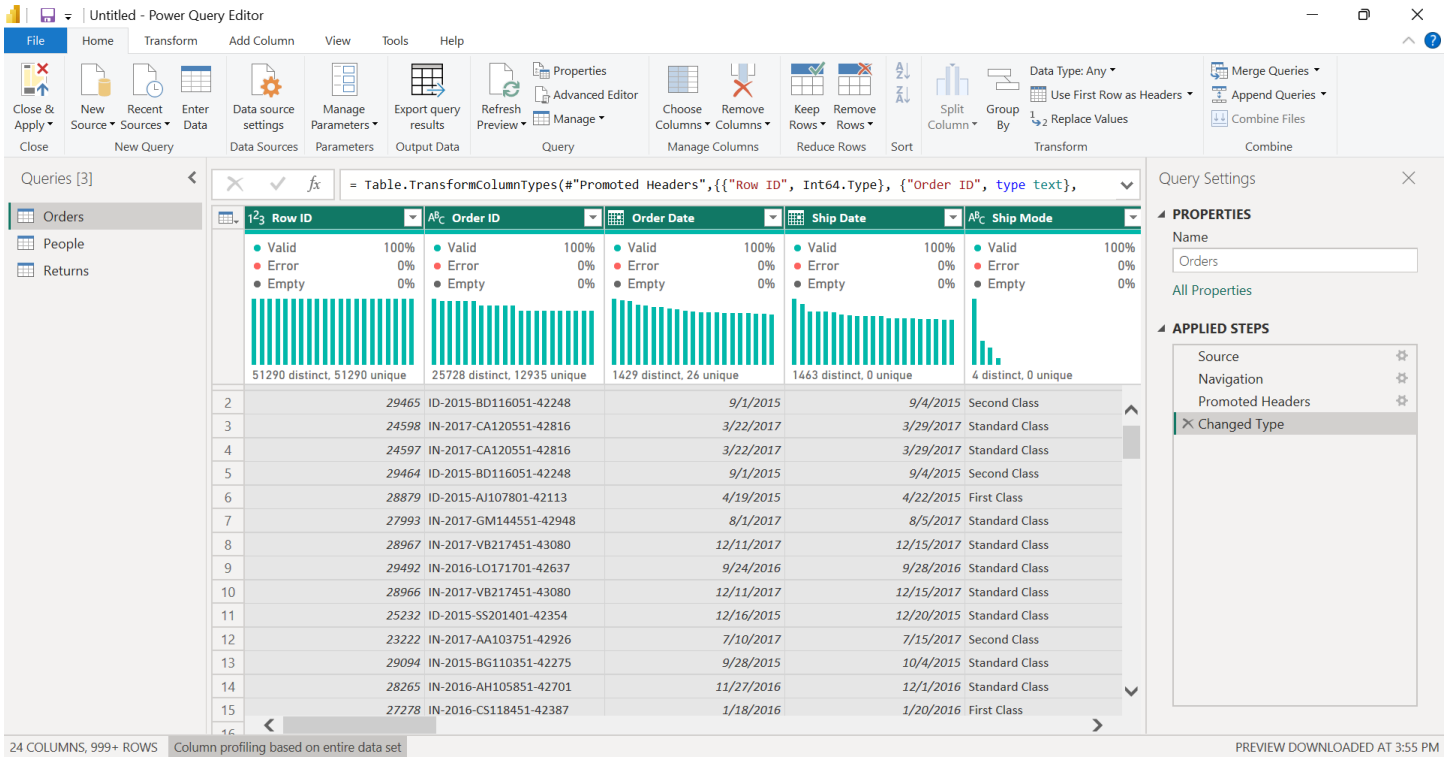
View tab > Tick Column Profile

Select each column one at a time and observe the profiling statistics displayed below the data preview grid.

Step 5: Record profiling observations.

For each column in each dataset, observe and record:

- Data type and total row count.



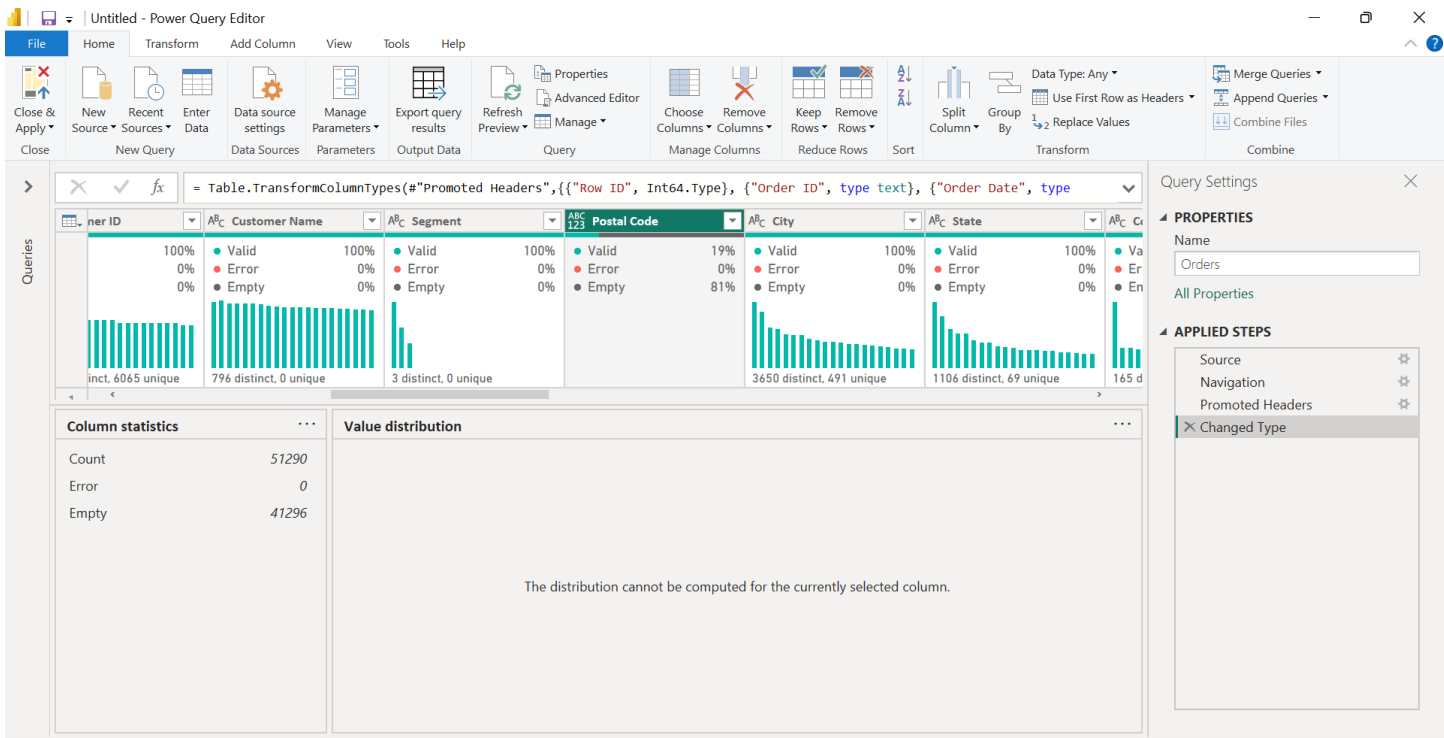
- Null/empty percentage from Column Quality.
- Distinct and unique value counts from Column Distribution.
- Minimum, maximum, mean, and standard deviation for numeric columns from Column Profile.
- Any errors, mismatched data types, or unexpected values.

Part C: Identify and Document Data Quality Issues

Step 6: Identify data quality issues from the profiling results.

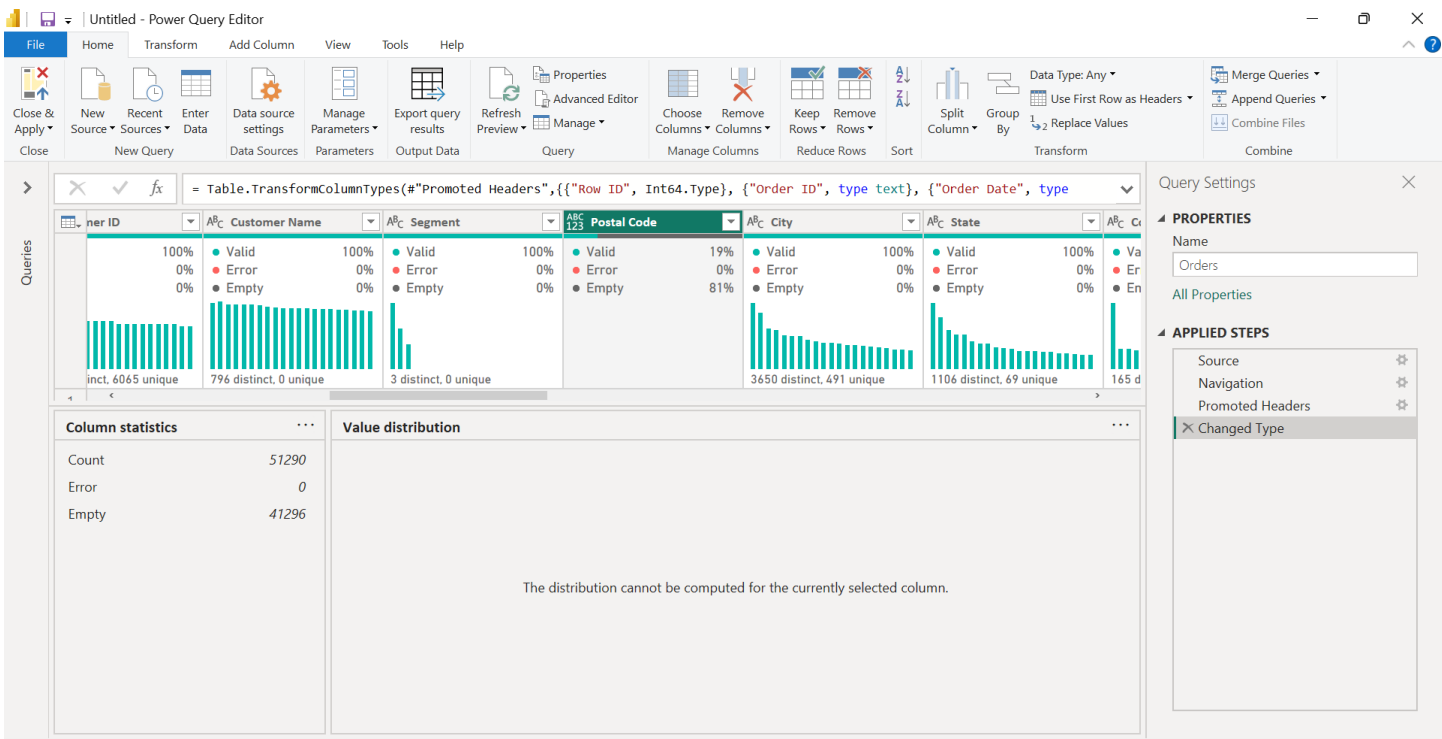
Check for common issues such as:

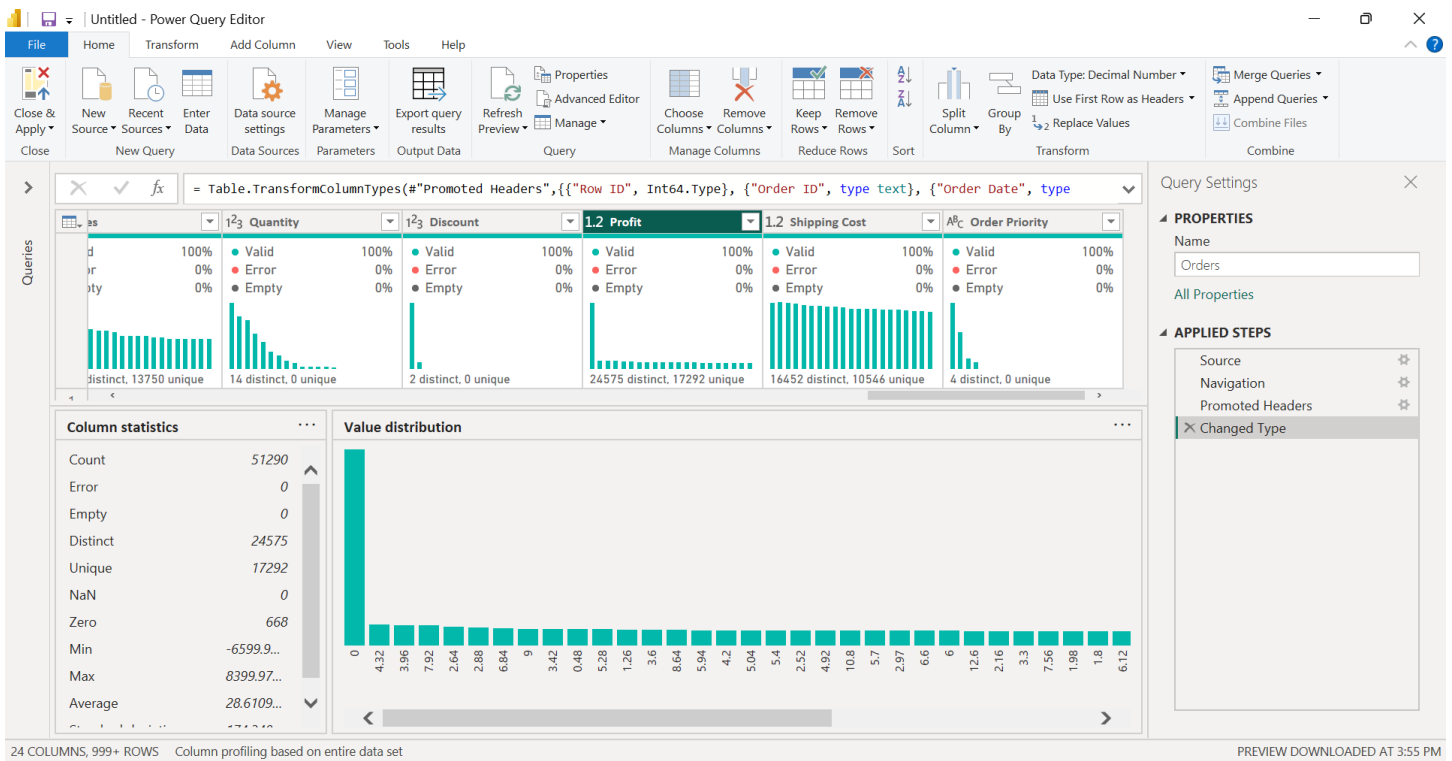
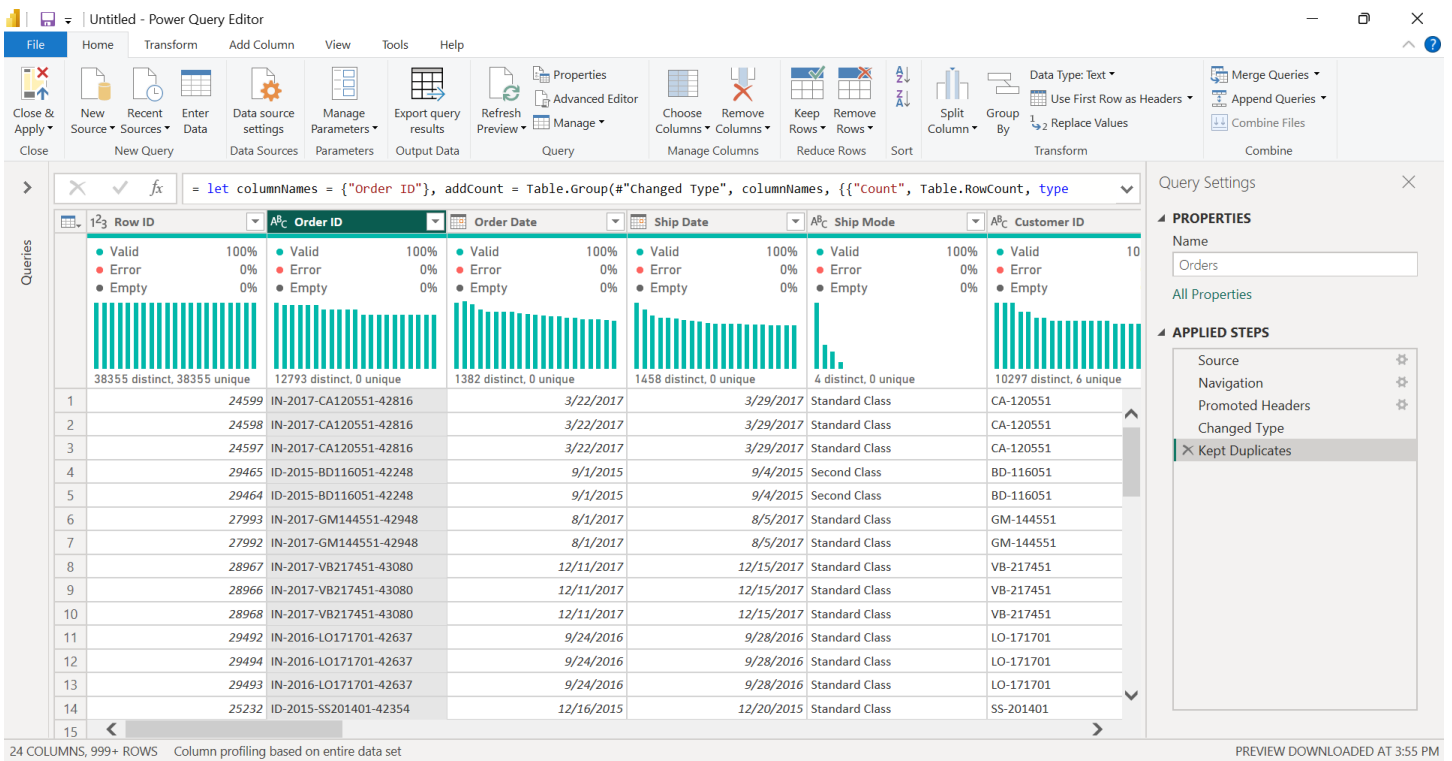
- High null or empty values in important columns.
- Columns containing only one unique value.
- Numeric columns containing text values or incorrect data types.
- Date values outside the expected range or future dates.
- Duplicate records indicated by low uniqueness.



Step 7: Document the identified issues.

Record at least three data quality issues found during profiling and mention a suitable remediation action for each issue.





7. RESULT

The Global Superstore 2018.csv dataset was successfully imported into Microsoft Power BI. Data profiling was performed using Power Query Editor with Column Quality, Column Distribution, and Column Profile. The dataset was examined for valid, empty, and error values, distinct and unique values, data types, and

statistical information. Data quality issues such as missing values, duplicate values, incorrect data types, and inconsistent entries were identified and documented for further data cleaning and analysis.

8. LEARNING OUTCOMES

- Understood the fundamental concepts of Business Intelligence and data acquisition.
- Gained practical knowledge of acquiring data from CSV, Excel, and SQL Server sources.
- Learned how to import and integrate heterogeneous datasets into Microsoft Power BI.
- Gained hands-on experience with Microsoft Power BI Desktop and Power Query Editor.
- Learned how to use Column Quality to evaluate valid, empty, and error values in datasets.
- Learned how to use Column Distribution to examine distinct and unique values and their frequency.
- Learned how to use Column Profile to examine detailed statistics and characteristics of individual columns.
- Developed the ability to identify missing values, duplicate records, incorrect data types, and inconsistent or unexpected values.
- Understood the importance of data profiling for evaluating data quality before analysis and visualization.
- Gained practical experience in documenting data quality issues and suggesting appropriate remediation actions.
- Developed a better understanding of how clean and reliable data supports accurate business analysis and decision-making.